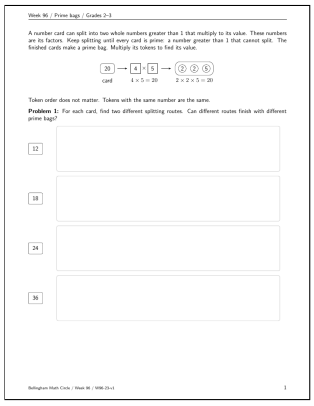
# Week 96 / Prime bags / Problem menu

for adults: pick, then print


Week 96 / Prime bags / Grades 2–3

A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 that cannot split. The finished cards make a prime bag. Multiply its tokens to find its value.

20 → 4 · 5 → 2 · 2 · 5

card: 4 × 5 = 20; 2 × 2 × 5 = 20

Token order does not matter. Tokens with the same number are the same.

**Problem 1:** For each card, find two different splitting routes. Can different routes finish with different prime bags?

12

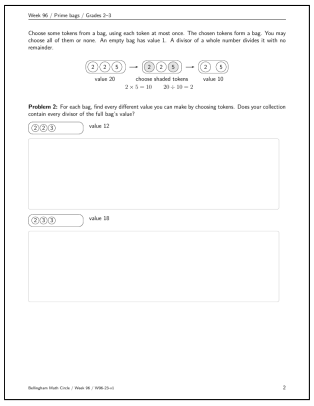
18

24

36

Bellingham Math Circle / Week 96 / W96-23-v1 1

**Different splitting routes, one prime bag?**
Grades 2–3 P1 · Grades 4–5 P1
first route
*wording differs by band*


Week 96 / Prime bags / Grades 2–3

Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.

2 · 2 · 5 → 2 · 2 · 5 → 2 · 5

value 20; choose shaded tokens; value 10

2 × 5 = 10; 20 ÷ 10 = 2

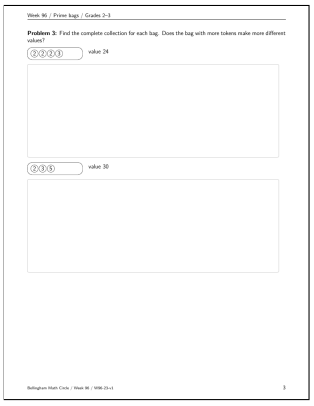
**Problem 2:** For each bag, find every different value you can make by choosing tokens. Does your collection contain every divisor of the full bag's value?

2 · 2 · 3 value 12

2 · 3 · 5 value 18

Bellingham Math Circle / Week 96 / W96-23-v1 2

**Every value a bag can make**
Grades 2–3 P2 · Grades 4–5 P2
first route
*wording differs by band*


Week 96 / Prime bags / Grades 2–3

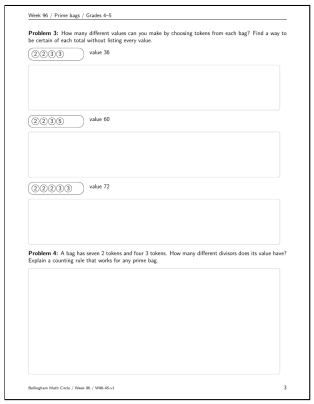
**Problem 3:** Find the complete collection for each bag. Does the bag with more tokens make more different values?

2 · 2 · 3 · 3 value 24

2 · 3 · 5 value 30

Bellingham Math Circle / Week 96 / W96-23-v1 3

**Four tokens against three**
Grades 2–3 P3
first route


Week 96 / Prime bags / Grades 4–5

**Problem 3:** How many different values can you make by choosing tokens from each bag? Find a way to be certain of each total without listing every value.

2 · 2 · 3 · 3 value 36

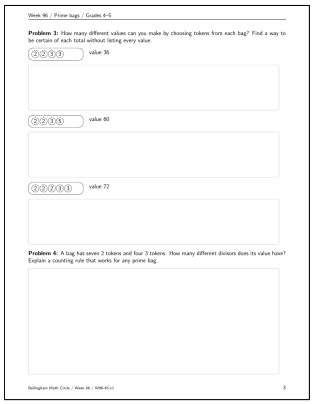
2 · 2 · 3 · 5 value 60

2 · 2 · 2 · 3 · 3 value 72

**Problem 4:** A bag has seven 2 tokens and four 3 tokens. How many different divisors does its value have? Explain a counting rule that works for any prime bag.

Bellingham Math Circle / Week 96 / W96-45-v1 3

**Count the values without listing them**
Grades 4–5 P3
first route


Week 96 / Prime bags / Grades 4–5

**Problem 3:** How many different values can you make by choosing tokens from each bag? Find a way to be certain of each total without listing every value.

2 · 2 · 3 · 3 value 36

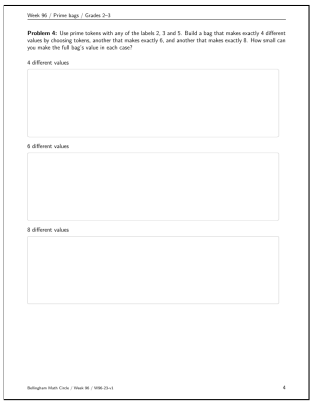
2 · 2 · 3 · 5 value 60

2 · 2 · 2 · 3 · 3 value 72

**Problem 4:** A bag has seven 2 tokens and four 3 tokens. How many different divisors does its value have? Explain a counting rule that works for any prime bag.

Bellingham Math Circle / Week 96 / W96-45-v1 3

**Seven 2s and four 3s**
Grades 4–5 P4
first route


Week 96 / Prime bags / Grades 2–3

**Problem 4:** Use prime tokens with any of the labels 2, 3 and 5. Build a bag that makes exactly 4 different values by choosing tokens, another that makes exactly 6, and another that makes exactly 8. How small can you make the full bag's value in each case?

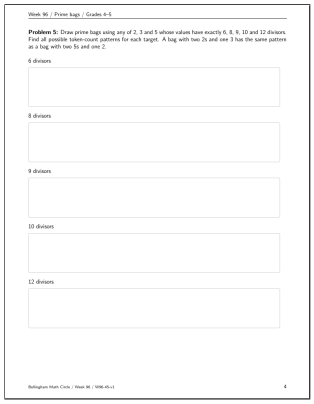
4 different values

6 different values

8 different values

Bellingham Math Circle / Week 96 / W96-23-v1 4

**Bags that make exactly 4, 6 or 8 values**
Grades 2–3 P4
return route


Week 96 / Prime bags / Grades 4–5

**Problem 5:** Draw prime bags using any of 2, 3 and 5 whose values have exactly 6, 8, 9, 10 and 12 divisors. Find all possible token-count patterns for each target. A bag with two 2s and one 3 has the same pattern as a bag with two 5s and one 2.

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors

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**Token-count patterns for a divisor count**
Grades 4–5 P5
return route


Week 96 / Prime bags / Grades 4–5

**Problem 6:** Find the least positive number with each divisor count below. You may use any primes. Explain how you can rule out every smaller number.

4 divisors

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors

Bellingham Math Circle / Week 96 / W96-45-v1 5

**The least number with each divisor count**
Grades 4–5 P6
return route